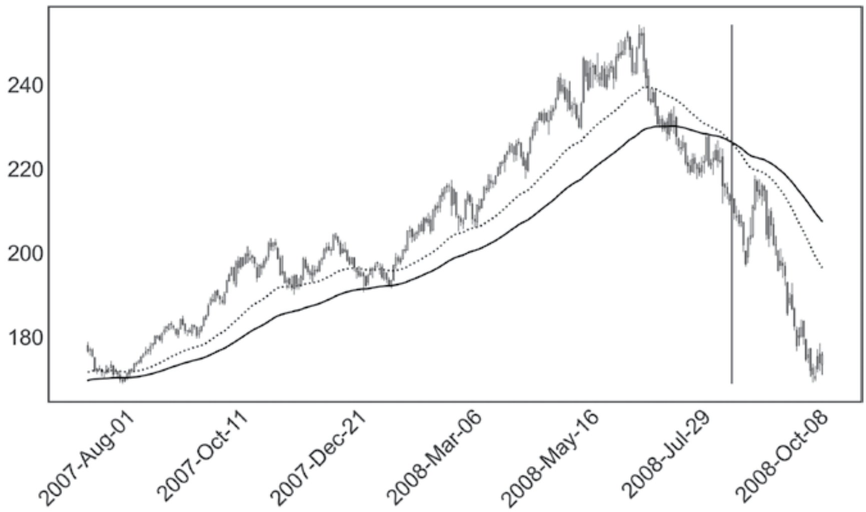

FIGURE 4.8 **Using moving average as trend filter.**

what market direction is dominating. If we set a criterion for our strategy that we are only allowed to buy when the overall market is in an uptrend and vice versa, we are likely to get two effects: the number of trades is reduced and there is a higher percentage of winners.

In Table 4.4, with data starting in 1992, we can see that the standard breakout strategy has significantly improved by using a simple trend filter. The annualised returns are up while the maximum drawdown is down and we got a few more positive months. Most importantly, the maximum drawdown has decreased significantly, and a much improved Sharpe ratio.

TABLE 4.4 **Adding a trend filter**

	Core model, combining the two	Breakout	MA Cross
Compounded annual return	15.81	14.25	12.74
Worst drawdown	−39.43	−47.19	−64.67
Annualised volatility	25.90	27.46	30.85
Percentage profitable months	57.08	52.92	54.17
Best month	37.04	48.70	37.94
Worst month	−18.37	−19.29	−21.04
Sharpe (zero)	0.70	0.62	0.54